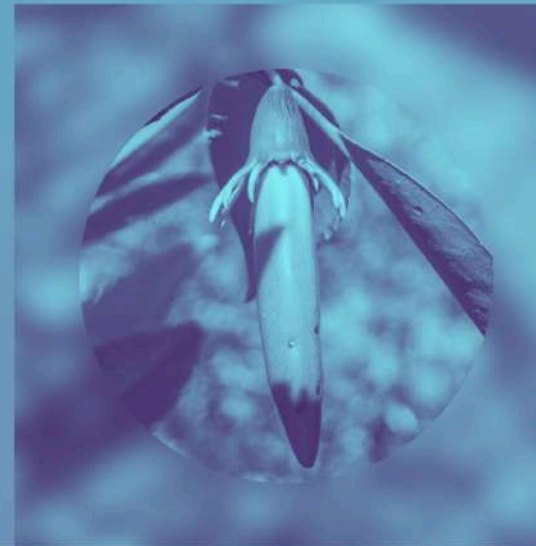


# B C A F

## PROJECT IMPACT REPORT



# Project in a Snapshot

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>                     | Community Resilience Through Blue Carbon Framework in Indonesia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Country</b>                   | Indonesia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Project Developer</b>         | Rekam Nusantara Foundation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Vision</b>                    | Capturing and preserving the extraordinary natural heritage of the archipelago to inspire collective awareness and action for environmental sustainability and a better future of Indonesia.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Overall Project Goals</b>     | Enhancing the management and governance of blue carbon ecosystems through monitoring, conservation, and rehabilitation efforts, while ensuring that the benefits of these ecosystems are accessible to coastal communities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>BCAF Supported Activities</b> | <ul style="list-style-type: none"><li>• Blue carbon ecosystem accounting</li><li>• Ecosystem restoration and rehabilitation</li><li>• Support community livelihood through capacity building</li><li>• Improvement of governance in blue carbon ecosystem conservation</li><li>• Education and enhancement of community awareness</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Main stakeholders</b>         | <ul style="list-style-type: none"><li>• Ministry of Marine Affairs and Fisheries (MMAF)</li><li>• Rekam Nusantara Foundation</li><li>• Konservasi Indonesia</li><li>• USAID Kolektif (which is collaboratively executed by Yayasan Konservasi Alam Nusantara and Yayasan Kehati)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>BCAF Focus</b>                | Support international projects to secure progress on blue carbon, coastal conservation and ocean accounting activities globally managed and coordinated by IUCN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>BCAF Funding Duration</b>     | 2023 – 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Type of Funding</b>           | Grant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Links/Synergies</b>           | <ul style="list-style-type: none"><li>• Indonesia's Updated Nationally Determined Contribution (NDC) for a Climate Resilient Future, as well as the Long-Term Strategy for Low Carbon and Climate Resilience 2050;</li><li>• Indonesian Biodiversity Strategy and Action Plan;</li><li>• SDGs National Action Plan 2021–2024, as well as Central Java and West Nusa Tenggara Provincial SDGs Action Plan;</li><li>• Presidential Regulation No. 98/2021 on the Implementation of Carbon Economy Value to Achieve National Contribution and Greenhouse Gas Emission Reduction;</li><li>• Management Strategy for Blue Carbon Ecosystem in the Blue Economy Development (MMAF)</li><li>• Central Java Governor Regulation No. 24/2019 on the Management Strategy for Mangrove Ecosystem;</li><li>• West Nusa Tenggara Governor Regulation No. 55/2023 on Sustainable Management of Snapper and Grouper Fisheries.</li></ul> |
| <b>BCAF Blue Print</b>           | Sustaining Mangrove Ecosystems A Holistic Approach to Conservation and Economic Growth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |





## Project Summary

**CENTRAL JAVA**

**WEST NUSA  
TENGGARA**

The project aims to strengthen the resilience of coastal communities by improving the management and governance of blue carbon ecosystems through monitoring, conservation, and rehabilitation efforts. It also seeks to ensure that the benefits of these ecosystems are accessible to local communities. Implemented in two key locations—Central Java and West Nusa Tenggara (Saleh Bay)—the initiative focuses on blue carbon asset inventory using the ocean accounts approach, ecosystem restoration and protection, and capacity building for alternative livelihoods.



Additionally, the project promotes inclusive governance by enhancing community participation in ecosystem conservation. By integrating ecological sustainability with socioeconomic development, it contributes to climate resilience and supports long-term well-being for coastal populations.



# Biodiversity and Natural Benefits

## West Nusa Tenggara

5 Plots



There were 9 seagrass species found in Saleh Bay. The total seagrass cover ranges from 5 – 90 %, with its density at Saleh Bay is 235.34 Ind/m<sup>2</sup>.

5 Plots



11 mangrove species were found in Sumbawa District. Mangrove community structure showed very good density of 3,651.39 Ind/ha, but has a low canopy cover of 31.75%.

Seagrass data collection plot

Mangrove data collection plot

There are 32 species of mangroves and 47 species of associated plants spread across the coast of Central Java and West Nusa Tenggara.

CO<sub>2</sub>

### AVERAGE CARBON STOCK VALUE

Central Java  
MangroveWest Nusa Tenggara  
Saleh Bay

| South Coast      | North Coast     | Mangrove                     | Seagrass         |
|------------------|-----------------|------------------------------|------------------|
| 116.68<br>MgC/ha | 77.61<br>MgC/ha | 463.95 ±<br>368.51<br>MgC/ha | 5,894.98<br>tonC |

## Central Java

### Demak

Demak district has 5 mangrove species, below are the mangrove condition at Demak

| Category           | 2023    | 2025    |
|--------------------|---------|---------|
| Mangrove Area (ha) | 1947.43 | 1782.32 |
| Density (Ind/ha)   | 3100    | 7733.33 |
| Canopy Cover (%)   | 72.83   | 70.4    |

### Jepara

Jepara district has 2 mangrove species found. Below are the mangroves condition

| Category           | 2023  | 2025    |
|--------------------|-------|---------|
| Mangrove Area (ha) | 91.78 | 118.79  |
| Density (Ind/ha)   | 4400  | 7333.33 |
| Canopy Cover (%)   | 57.4  | 49.61   |

### Cilacap

Cilacap district has 8 mangrove species, below are the mangrove condition at Cilacap

| Category           | 2023    | 2025    |
|--------------------|---------|---------|
| Mangrove Area (ha) | 8477.96 | 8850.24 |
| Density (Ind/ha)   | 1280    | 4850    |
| Canopy Cover (%)   | 53      | 64.23   |

### Kebumen

Kebumen district has 5 mangrove species, below are the mangrove condition at Kebumen

| Category           | 2023  | 2025    |
|--------------------|-------|---------|
| Mangrove Area (ha) | 16.19 | 48.89   |
| Density (Ind/ha)   | 4170  | 4033.33 |
| Canopy Cover (%)   | 61.66 | 67.67   |

15 Plots

Mangrove  
on 4 District





## Marine Protected Areas

The project supported protection and adaptive management in five Marine Protected Areas (MPAs) in Central Java and West Nusa Tenggara, Indonesia.

All MPAs include at least one key blue carbon habitat (mangrove, seagrass) and have been assessed to measure the management effectiveness.

| MPA Name                                        | AREA (ha)                |              |                     | Blue Carbon Ecosystem | Management Effectiveness Score                                 |        |
|-------------------------------------------------|--------------------------|--------------|---------------------|-----------------------|----------------------------------------------------------------|--------|
| Panjang Island, Jepara, Central Java            | TOTAL AREA: 494.25       |              |                     | Seagrass, Mangrove    | 2023                                                           | 2024   |
|                                                 | Core Zone                | Limited Zone | Rehabilitation Zone |                       | 13.95%                                                         | 41.51% |
|                                                 | 10.44                    | 479.52       | 4.29                |                       | Improved Effectiveness Score                                   |        |
| Betahwalang, Demak, Central Java                | TOTAL AREA: 244.88       |              |                     | Mangrove              | N/A                                                            |        |
|                                                 | Core Zone                | Limited Zone | Rehabilitation Zone |                       | Established on May 2024, does not have Effectiveness Score yet |        |
|                                                 | 33.44                    | 211.44       | N/A                 |                       |                                                                |        |
| Karang Jahe, Rembang, Central Java              | TOTAL AREA: 33.60        |              |                     | Seagrass, Mangrove    | 2023                                                           | 2024   |
|                                                 | Core Zone                | Limited Zone | Rehabilitation Zone |                       | 13.94%                                                         | 45.13% |
|                                                 | 1.37                     | 24.13        | 8.15                |                       | Improved Effectiveness Score                                   |        |
| Liang-Ngali Island, Sumbawa, West Nusa Tenggara | TOTAL AREA: 32,644.43    |              |                     | Seagrass, Mangrove    | 2023                                                           | 2024   |
|                                                 | Core Zone                | Limited Zone | Rehabilitation Zone |                       | 57.57%                                                         | 71.64% |
|                                                 | 1,405.54                 | 29,052.15    | 757.92              |                       | Improved Effectiveness Score                                   |        |
| Lipan-Rakit Island, Sumbawa, West Nusa Tenggara | TOTAL AREA: 26,670.80 ha |              |                     | Seagrass, Mangrove    | 2023                                                           | 2024   |
|                                                 | Core Zone                | Limited Zone | Rehabilitation Zone |                       | Established                                                    | 50.18% |
|                                                 | 991.52                   | 25,621.17    | 58.11               |                       | Optimally Managed                                              |        |



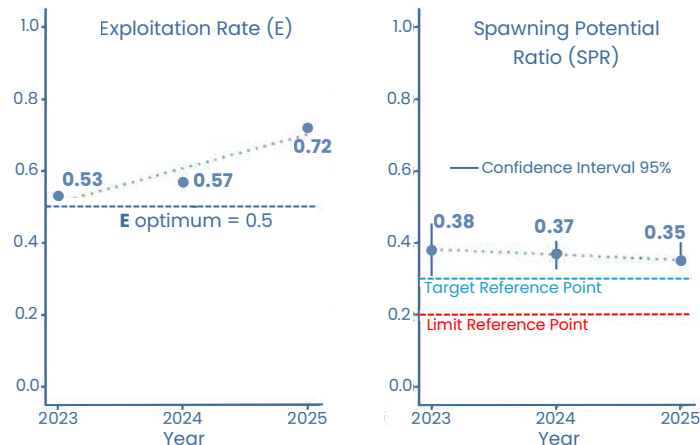
## Central Java

- Landing sites on the north coast of Central Java are at Betahwalang, Teluk Awur, and Wedung.
- Data collected since May 2023 – May 2025 on North Coast of Central Java can be seen below:



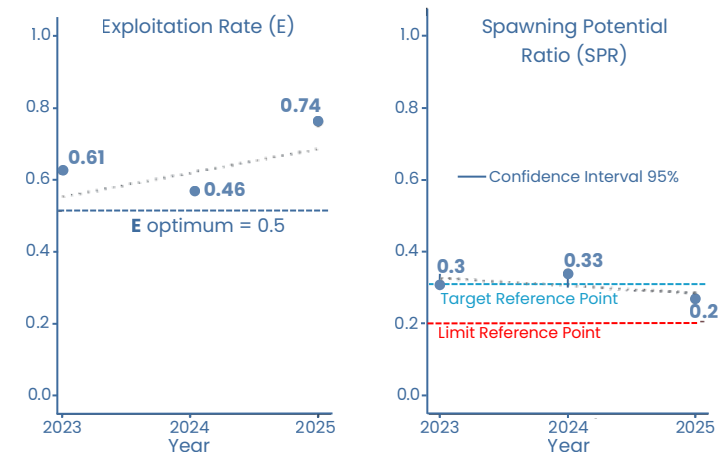
### *Portunus pelagicus* (Blue Swimming Crab) at Jepara

- Total Trips: 153
- Length Measurements: 2,535



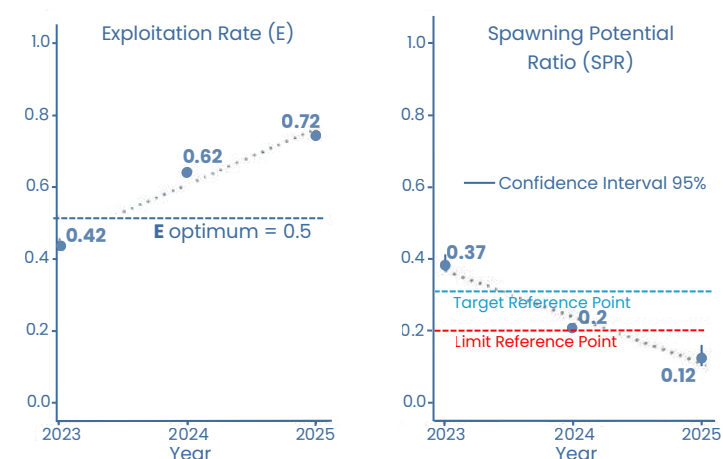
### *Portunus pelagicus* (Blue Swimming Crab) at Demak

- Total Trips: 301
- Length Measurements: 7,489



### *Scylla serrata* (Giant Mud Crab) at Demak

- Total Trips: 941
- Length Measurements: 7,229



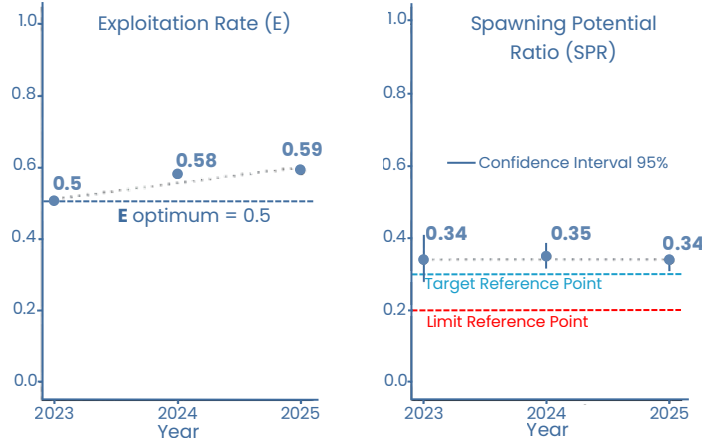
## West Nusa Tenggara

- Landing sites in Saleh Bay are at Batu Bangka, Labuhan Kuris, Labuhan Sangoro, Labuhan Bontong and Teluk Santong.
- Data collected since May 2023 – May 2025 on Saleh Bay can be seen below:



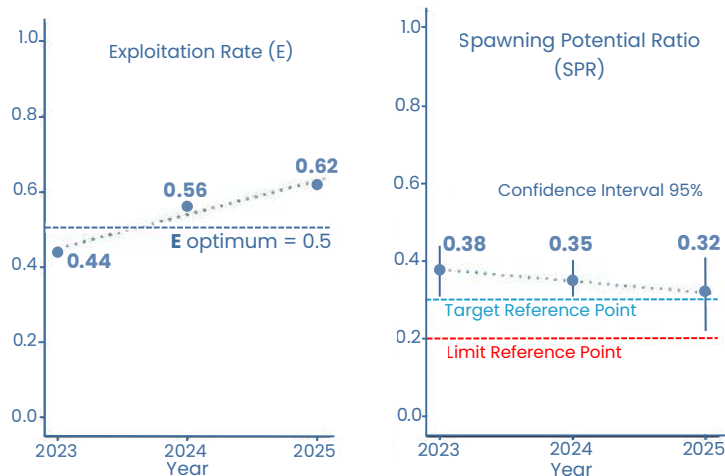
### *Portunus pelagicus* (Blue Swimming Crab)

- Total Trips: 1,259
- Length Measurements: 3,298



### *Siganus canaliculatus* (Rabbitfish)

- Total Trips: 823
- Length Measurements: 1,839



From May 2023 – May 2025, the total number of data recorded:



**7**  
Enumerators



**3,581**  
Trip Data



**22,501**  
Length Measurements

in the north coast of Central Java and Saleh Bay







# Rehabilitation and Restoration

## Central Java

|                                                                                   | Rehabilitation                          | Survival Rate      |
|-----------------------------------------------------------------------------------|-----------------------------------------|--------------------|
|  | Mangrove:<br>149,500 seeds<br>( 15 ha ) | Mangrove:<br>79.8% |
|  | Seagrass:<br>500 seeds<br>( 500 m2 )    | Seagrass:<br>50%   |

## West Nusa Tenggara

|                                                                                   | Rehabilitation                       | Survival Rate    |
|-----------------------------------------------------------------------------------|--------------------------------------|------------------|
|  | Mangrove:<br>9,500 seeds<br>( 5 ha ) | Mangrove:<br>90% |
|  | Seagrass:<br>2000 seeds<br>( 2 ha )  | Seagrass:<br>30% |

## Central Java

- Participatory mapping was conducted as an effort to increase the effectiveness and accuracy of the mangrove planting program using Google Earth Engine.
- The determination of mangrove planting sites in Central Java involved an in-depth analysis of the ecological, hydrological, topographical, and socio-economic conditions of the areas to be restored.

## Saleh Bay

- Rehabilitation of *Rhizophora mucronata* seedlings held in Labuhan Kuris Village at October 2024 and January 2025
- Rehabilitation of *Enhalus acoroides* seedlings held in Labuhan Jambu Village on May 2024 showed a low survival rate of about 30% due to seedlings buried in sand from high sedimentation which inhibited photosynthesis.





## CENTRAL JAVA

6

Groups

Supported



13  
Coastal Community  
Groups

7

Groups

## WEST NUSA TENGGARA



±105

Total Group Members

±120



Total Group Members

*Empowering Community-based Business that Based on Blue Carbon*



Eco-tourism  
Development



Mangrove-based  
Silvofishery



Crab Hatchery  
and Farming



Mangrove  
Seedling  
Nurseries



Vegetable  
Farming



Seagrass and  
Mangrove  
Rehabilitation



Fisheries and  
Fish Product  
Processing

### What We Empower

Since January 2024,  
a total of:

15

Workshop and  
training held at  
Central Java

The support included technical trainings on seagrass and mangrove rehabilitation, fish processing, aquaculture innovation, community-based monitoring, eco-tourism development, and financial management. Groups received essential tools and infrastructure such as crab hatchery and crab house systems, processing equipment, nursery facilities, farming gear, and administrative materials.

Legal assistance was also provided, particularly to help secure land tenure through KHDPK permits, which strengthened community ownership and governance. In addition, the team facilitated awareness-building through workshops, field mentoring, public campaigns, and cross-sector partnerships that nurtured sustainable, community-led coastal stewardship.

Since December 2023,  
a total of:

7

Workshop and  
training held at  
West Nusa Tenggara

### Most Notable Impact

#### Central Java

##### 1. Crab Farming Training - Jepara

- Attended by 15 participants, it inspired them to explore and develop various innovative business models.
- Several groups began applying crabhouse methods, fattening techniques using gallons, and soft-shell crab farming.
- The result of the training was an increase in harvest frequency, harvest yield, and individual weight gain of mangrove crabs, from 100-150 grams to 200-250 grams each.
- It successfully established a crab hatchery unit by the Kolak Sekancil group, a self-initiated effort that had never existed before.

##### 2. Catatan Cuan - Cilacap

- Catatan Cuan is an educational board game designed to help micro, small, and medium enterprises (MSMEs) learn how to manage their finances in an interactive and enjoyable way.
- Attended by 24 participants, it successfully empowered them to understand key concepts in economic and natural resources management.
- It also promotes administrative skills and financial management.

#### West Nusa Tenggara

##### 1. Sambal Rebon Masin Training

- Attended by 20 participants from Saro Mase group to develop sambal rebon masin, a traditional Sumbawa condiment made from mangrove-sourced rebon.
- The group learned hygienic production, packaging, labeling, and basic marketing, also assistance in obtaining product permits, and assistance in preparing business feasibility documents.
- They successfully made a bottled, ready-to-sell product, improved their skills in value-added processing, and strengthened the identity of a local specialty with tourism potential.

##### 2. Seagrass Rehabilitation & Monitoring Training

- Collaboration between the community, government, universities, and other stakeholders in preserving the coastal environment, especially blue carbon ecosystems.
- 7 participants learned simple rehabilitation and monitoring techniques tailored to local conditions.
- The training sparked active involvement in seagrass monitoring, especially in areas they helped restore.
- Successfully rehabilitated 1,150 seagrass seedlings.





## Outreach and Awareness

### World Mangrove Day 2024

On July 2024, we hosted a series of community events across Cilacap, Demak, and Kebumen to promote mangrove conservation, public awareness, coastal clean-ups, and children's coloring competition.



attended by  
**260**  
participants



planted  
**4000**  
mangrove  
seedlings



### Talkshow and Movie Screening

On November 2024, a documentary on mangrove ecosystem services and their connection with local communities was produced and screened during a public talkshow series across four districts of Central Java, namely Cilacap, Kebumen, Demak, and Jepara.



attended by  
**309**  
participants

### Education and Campaigns

On February and March 2025, we conducted blue carbon ecosystem awareness campaigns to future generations, utilizing 4 interactive modules at Central Java and West Nusa Tenggara

Held at

**3**

school, with a total of

**147**

students



### The Ocean's Tale of Nusantara

Held on May 2025, at Jakarta, The Ocean's Tale of Nusantara was a powerful public event showcasing stories of coastal communities through visual storytelling, 7 movie screenings, exclusive blue carbon documentary movie launch, and public dialogue.



attended by  
**300**  
people



## SOCIAL MEDIA REPORT



Instagram

**162,823**  
Reach

**246,961**  
Views

**4,820**  
Like

**28**  
Comment

**371**  
Share

**360**  
Save



LinkedIn

**11,628**  
Impression

**4,166**  
Views

**245**  
Reaction

**2**  
Comment

**50**  
Repost

**23.52%**  
Engagement Rate

**25**

Contents  
Made



**FRCI**  
FISHERIES RESOURCE  
CENTER OF INDONESIA



Instagram

**13,181**  
Reach

**19,960**  
Views

**1,118**  
Like

**9**  
Comment

**47**  
Share

**91**  
Save



LinkedIn

**39,826**  
Impression

**8,211**  
Views

**864**  
Reaction

**5**  
Comment

**69**  
Repost

**18.58%**  
Engagement Rate

**47**

Contents  
Made

## Governance, Policy, and Financial

### Development of Sikajeng Application

- Sikajeng is a digital app developed by Environmental and Forestry Office (DLHK) of Central Java to collect and map mangrove rehabilitation and restoration data across the province.
- It functions as a central tool to record, monitor, and visualize mangrove ecosystem recovery for better coordination and decision-making.
- Used collaboratively by conservation groups, fisherfolk, community patrols, forestry and fisheries officers, DLHK, DKP, BAPPEDA, and local planning bodies.
- Supports a collaborative, bottom-up approach to ecosystem restoration and improves stakeholder coordination at the provincial level.



### Community Forest Initiatives

- The Special Forest Management Area (KHDPK) is an initiative by Ministry of Environment and Forestry (KLHK) to grant communities legal access for sustainable forest management and resolve land tenure conflicts.
- Empowers local groups to manage forest areas through social forestry schemes like Community Forestry (HKM) and Village Forests.
- Involves KTH Sida Asih, LMDH Bejagan, Perhutani, CDK, BPSKL-KLHK Yogyakarta, and local authorities through FGDs and workshops.
- Promotes equitable forest governance by enabling local management of 123 hectares for mangrove conservation, ecotourism, aquaculture, seedling production, and agriculture. Currently in the proposal submission stage to KLHK.

### Blue Carbon Document Initiatives

#### Prospectus Document for Blue Carbon in Marine Protected Areas

A document that provides an overview of the potential for blue carbon in Indonesia. Together with MMAF, KI, and USAID Kolektif, we participated in the drafting of the document

#### Practical Guidelines for the Inventory of Greenhouse Gas Emissions in MPAs

This document can be used as a technical guide for blue carbon data collection and processing methods to produce standardized national data.

**DRAM :**  
Mitigation Action Design Document

**RAM :**  
Mitigation Plan of Action Document







## Success and Challenges

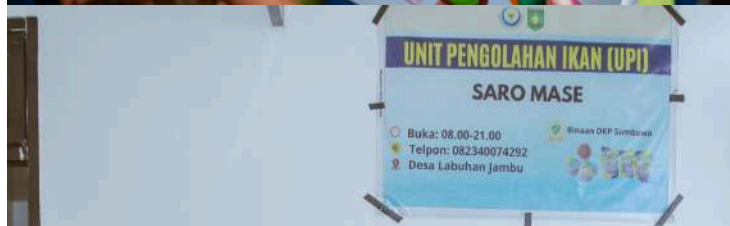
Blue carbon ecosystem management in Central Java and Saleh Bay faces several challenges. One key issue is changing government leadership, which affects mangrove-related policies. At the same time, national budget efficiency efforts and ministerial restructuring delayed KHDPK applications. In response, the team worked to engage key stakeholders and promote sustainable mangrove management through media campaigns and community outreach. Partnerships with NGOs and private companies also helped. To manage limited funding, simple and adaptable technologies were used.

Environmental conditions like strong winds and high waves disrupted fishing activities, ecological surveys, and educational programs in Kampung Laut. Turbid waters also delayed underwater photography. To work around this, the team created a flexible schedule based on BMKG weather forecasts. Land-based photography was used when water conditions weren't ideal, and activity plans were adjusted to suit the weather. For mangrove planting in wave-exposed areas, the Relay Encased Method (REM) was used. In seagrass rehabilitation, low growth rates posed a challenge, requiring interventions to reduce sediment buildup. To support this, a pilot seagrass nursery using seagrass fruits was developed in collaboration with SEACREST Indonesia.

Support for blue carbon programs remains limited, particularly for seagrass conservation. Central Java's Bappeda assigned the program to DLHK, which could restrict broader collaboration. Budget allocations for 2025 are also only under DLHK. To strengthen this, planning and funding support from DKP is needed, along with forming a Seagrass Conservation Forum and developing a conservation plan matrix to foster multi-sector cooperation.

On the livelihood side, the marketing and distribution of sambal masin still face issues due to unclear market prospects and administrative barriers, such as group-based requirements from the Health Office. These can be addressed through partnerships with local universities and institutions, along with market studies to assess potential. Direct engagement with the Health Office is also crucial to facilitate on-site assessments.

In terms of seagrass conservation, regulatory language and the idea of creating protected zones remain sensitive topics. To address this, stakeholders agreed to develop village-based education and rehabilitation sites. Rather than using the term "conservation area", the location was framed as a "field education center" to make it more acceptable to communities and local authorities.







## Looking Beyond! (What's Next?)



## Biodiversity and Natural Benefits

- Continue monitoring carbon stock and ecosystem changes in Saleh Bay and coastal Java by involving multiple stakeholders.



## Rehabilitation and Restoration

- Expand rehabilitation areas using adaptive planting methods, with regular maintenance and monitoring of environmental changes.
- Rehabilitation efforts are guided by analysis of past data, including sedimentation, water quality, and social conditions.



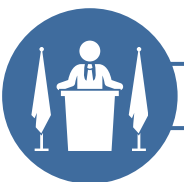
## Community and Livelihood

- Promote alternative livelihoods through the development of businesses based on Non-Timber Forest Products (NTFP).
- Promote ecotourism products and Non-Timber Forest Products (NTFP).
- Train communities in ecosystem carbon data collection and monitoring to foster citizen science practices.
- Food safety training for Sambal Masin producers to ensure the quality and safety of home-based food products.
- Expansion of Sambal Masin distribution to souvenir centers in Sumbawa Island and beyond, as well as to modern retail stores (e.g., Alfamart and Indomaret).
- Digital marketing of Sambal Masin through e-commerce platforms (e.g., Tokopedia and Shopee).
- Community capacity building in microfinance management.



## Outreach and Awareness

- Develop curriculum and provide teacher training for youth education on blue carbon ecosystems.
- Enhance digital and community engagement on blue carbon ecosystems through an e-learning platform.



## Governance, Policy, and Financial

- Strengthen multi-stakeholder collaboration in preserving blue carbon ecosystems.
- Establish sustainable financing mechanisms with the private sector for blue carbon conservation and management.





# BCAF PROJECT IMPACT REPORT

